Quiz 1

Name: _____

AI 501: Mathematics for AI | Duration: 20 min | Student ID: _____
Total: 10 marks

Question 1 (3 marks)

For each task listed in the table below, place a tick in exactly one of the three columns.

Task	Regression	Classification	Neither
(i) Predicting a student's cumulative percentage score in AI 501 at the end of term.	☐	☐	☐
(ii) Predicting a student's letter grade in AI 501 at the end of term.	☐	☐	☐
(iii) Predicting the probability that electricity prices in Pakistan decrease during the next year.	☐	☐	☐
(iv) Predicting whether electricity prices in Pakistan decrease during the next year.	☐	☐	☐
(v) Predicting the number of vehicles that cross a given intersection during the next hour, from camera footage.	☐	☐	☐
(vi) Grouping the enrolled students into cohorts of similar academic background, no cohort labels having been recorded.	☐	☐	☐

Solution:

Task	Regression	Classification	Neither
(i) Cumulative percentage score	✓		
(ii) Letter grade		✓	
(iii) Probability of a decrease	✓		
(iv) Whether prices decrease		✓	
(v) Number of vehicles in the next hour	✓		
(vi) Grouping students into cohorts			✓

Question 2 (3 marks)

For $v = (1, -1, 2)$, find the unit vector (a vector with 2-norm equal to 1) pointing in the *opposite* direction to v.

Solution:

$$\|v\|_2 = \sqrt{1^2 + (-1)^2 + 2^2} = \sqrt{6}, \qquad -\frac{v}{\|v\|_2} = \frac{1}{\sqrt{6}}(-1, 1, -2).$$

Question 3 (4 marks)

Suppose we are describing 3 apartments. For each apartment we record several numerical *features*, and each feature is written as a vector whose i-th entry is that feature's value for apartment i:

$$f_1 = (3, 2, 2), \qquad f_2 = (2, 1, 3), \qquad f_3 = (1, 2, -6),$$

A student claims that one of these three features is *redundant* and carries no new information beyond what the other two already capture, because it can be written as a linear combination of

them. Determine whether this is the case, and if so, identify the redundant feature and express it as a linear combination of the other two.

Solution: Seek a, b with $f_3 = af_1 + bf_2$. Coordinatewise,

$$3a + 2b = 1, \qquad 2a + b = 2, \qquad 2a + 3b = -6.$$

The first two give $a = 3$, $b = -4$; these satisfy the third, since $2(3) + 3(-4) = -6$. Hence

$$f_3 = 3f_1 - 4f_2,$$

so the claim holds: the features are linearly dependent and f_3 carries no information beyond f_1 and f_2.